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310404

Roll No. _____ Total No of Pages: 4

310404
B. Tech. III - Sem. (Main) Exam., February - 2021
Computer Science & Engineering
3CS3-04 Digital Electronics
Common for CS/ IT

Time: 3 Hours **Maximum Marks: 120**

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 10 × 2 marks = 20 marks. All ten questions are compulsory.*

Part – B: *Analytical/Problem Solving questions (up to 100 words) 5 × 8 marks = 40 marks. Candidates have to answer five questions out of seven.*

Part – C: *Descriptive/Analytical/Problem Solving questions 4 × 15 marks = 60 marks. Candidates have to answer four questions out of five.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

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PART - A

Q.1 $(10101011)_8 = (?)_{16}$ [2]

Q.2 $(110101)_{BCD} = (?)_{Excess\text{-}3}$ [2]

Q.3 Prove the following using Boolean algebra laws. x (x + y) = x [2]

Q.4 Differentiate between unipolar & bipolar logic family. [2]

Q.5 Which flip – flop can work as frequency divider & Why? [2]

Q.6 What is the fundamental difference between half adder and full adder? [2]

Q.7 What is the fundamental element of RAM? [2]

Q.8 Find the value of r $(54)_6 = (42)_r$ [2]

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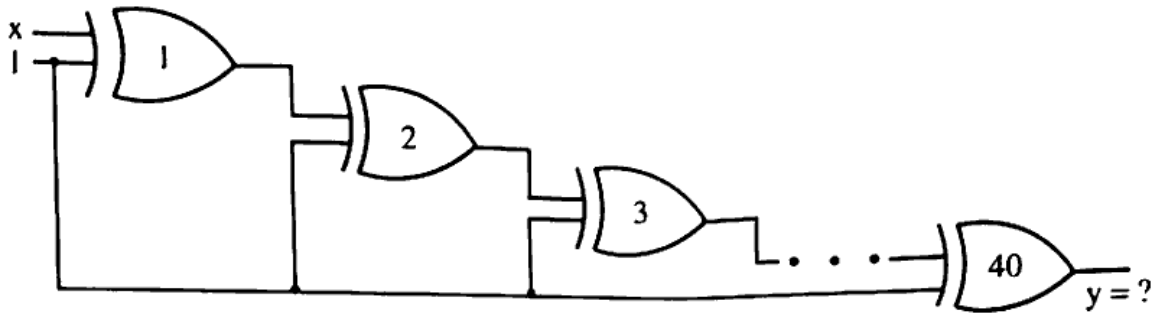
Q.9 What is the Self – complementing code? Explain with example. [2]

Q.10 What is Gray code? Why it is important? [2]

PART – B

Q.1 Explain staircase switching with truth table. Design circuit using basic logic gates and then design it using NAND gate only. Further reduce it upto 4 gates only. With proper procedure. (6 marks for circuit diagram and its minimization) [8]

Q.2



40 Ex – OR gates are arranged as shown in the figure. Find out the output after each gate as well as final gate for all the possible values of x. [8]

Q.3 Design serial adder with proper justification. https://www.btubikaner.com [8]

Q.4 Design full subtractor using 2:1 MUX. Also justify the truth table. [8]

Q.5 Design the following circuit using X – NOR gate (s) and NOR gate (s) only. (3 marks for truth table and justification, 1mark for X – NOR design and 4 marks for NOR design)

$$z = \begin{cases} \bar{x}, \text{if } y = 0 \\ x, \text{if } y = 1 \end{cases}$$ [8]

Q.6 (a) Prove the following using Boolean Algebra postulates and theorems. [4]

$AB + \bar{A}C = (A + C)(\bar{A} + B)$

(b) Minimize the following function using Karnaugh map - [4]

$f(w, x, y, z) = \sum_m (0, 2, 6, 8) + \sum_d (1, 3, 4, 10)$

Q.7 (a) Find y – x using (r – 1)'s complement where - [4]

$x = (101101)_2$, $y = (1000000)_2$

(b) Find x – y using r's complement where - [4]

$x = (1001101)_2$, $y = (1000000)_2$

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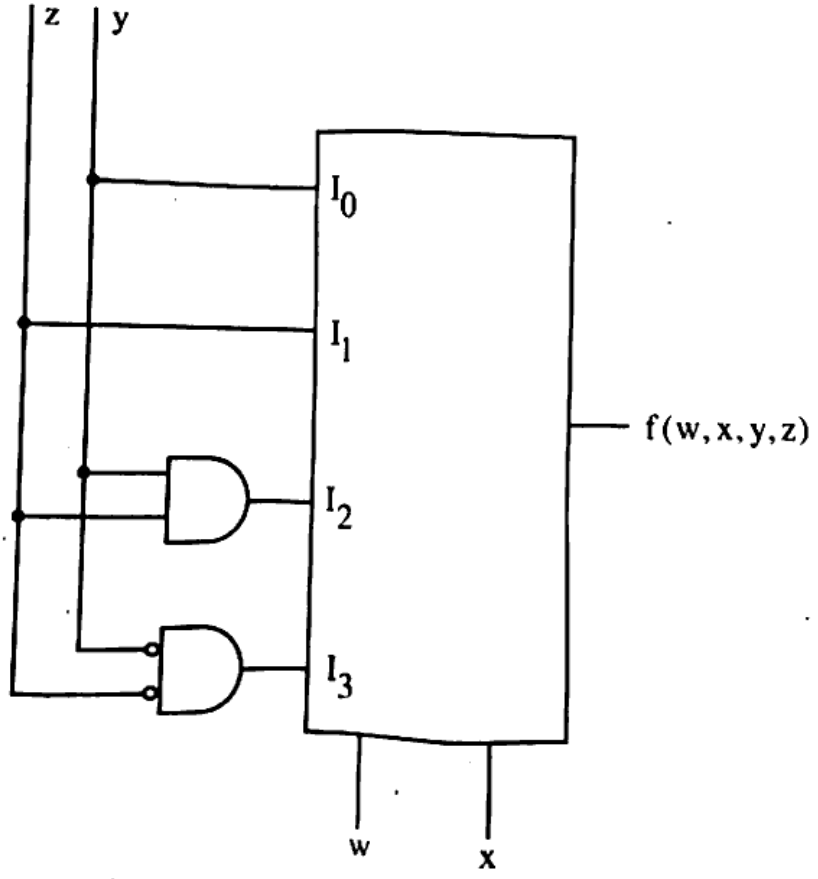
PART – C

Q.1 Minimize the following function using Quine McCluskey method - [15]

$F(a, b, c, d) = \sum_m (1, 4, 6, 7, 8, 9, 10, 11, 15)$

Q.2 What is the fundamental difference between Ring & Johnson counter? Explain with their diagrams and waveforms. [15]

Q.3 Find the Boolean function implemented by following MUX with proper justification [15]



Q.4 Is there any difference between TTL with active pull up and TTL with Totem – Pole output? Discuss in detail with the help of circuit diagram. (8 marks for diagram) [15]

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